In this study, we analyzed the evolutionary distances of 16,210 one-to-one orthologous genes between *Homo sapiens* and 87 other mammalian species, totaling 1,259,294 protein sequences. This large-scale analysis allowed us to identify genes and biological pathways that show the highest and lowest levels of divergence across different species to humans. Based on these patterns, we examined where immune-related genes fall within the total gene set. Our findings highlight specific genes and pathways that are more divergent, and therefore more likely to be species-specific with potential functional implications. We also identified species that accumulated more substitutions than others when compared to humans for these genes.

This study is based on a comparative genomic analysis between *Homo sapiens* and other mammalian species, using evolutionary distances estimated through phylogenetic methods. The main goal was to investigate evolutionary patterns in orthologous genes, particularly those related to the immune system. The analysis began with a comprehensive list of all human protein-coding genes, from which 1:1 orthologs were identified across 98 mammal species. Each gene was processed individually through an analysis pipeline, described in detail below, which generated evolutionary distance estimates between each species' gene sequence and the corresponding human sequence.

The analyses were structured in two main approaches: (i) an overall analysis across all genes and species, aiming to identify general divergence patterns. This included identifying genes with the highest and lowest levels of divergence and evaluating divergence across biological pathways; and (ii) targeted analyses focusing on a subset of species and the most divergent genes identified in the overall analysis. This included detecting signals of positive selection at specific sites and mapping divergence patterns within gene interaction networks.

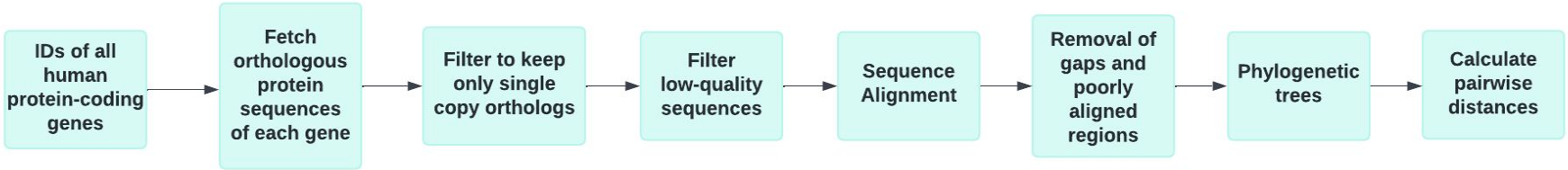


Figure 1: Flowchart from data collection to pairwise distance calculation.

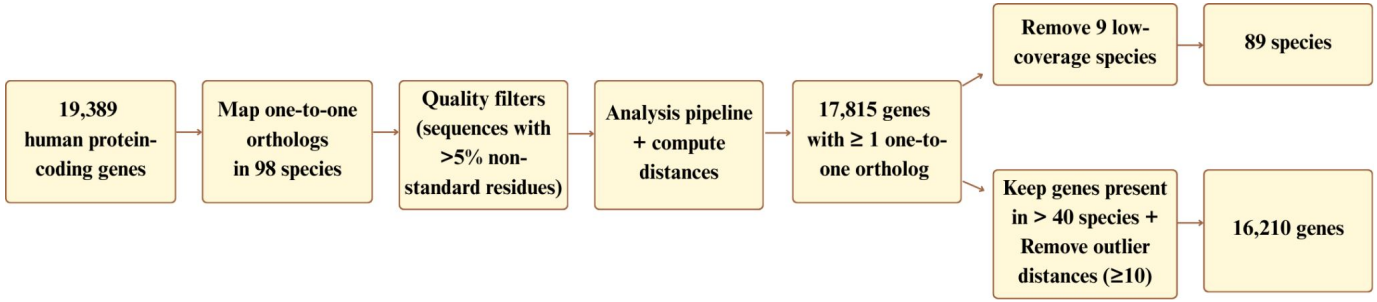


Figure 2: Flow of gene and species filtering. From 19,389 human protein-coding genes, one-to-one orthologs were mapped in 98 mammals. Sequences with >5% non-standard residues were removed and evolutionary distances were computed. After retaining genes with ≥1 ortholog (17,815), removing 9 low-coverage species (final species set: 89), keeping genes present in > 40 species, and excluding 68 distance outliers (distance ≥ 10), the final dataset comprised 16,210 genes.

Descriptive analysis of gene distances

Using data from the pipeline described above, a genetic distance database was created for 16,210 single-copy orthologous genes between *Homo sapiens* and 87 other mammalian species. From this database, the following analyses were performed to identify genes with the greatest evolutionary distances and to estimate the distances of immunological pathways based on the average genetic distance of their constituent genes.

For each gene, the median genetic distance was calculated, as the median is less affected by outliers and provides a more reliable estimate of overall evolutionary divergence. The median is appropriate in evolutionary studies since it offers a picture of typical genetic differences, especially when some species pairs are more divergent due to long evolutionary separation or rapid adaptation.

Genes were then ranked by their median genetic distances and grouped into percentiles. The 15th percentile (most conserved genes) and the 85th percentile (most divergent genes) were identified. This categorization helped highlight genes with distinct evolutionary patterns, allowing for further analysis of their functional significance.

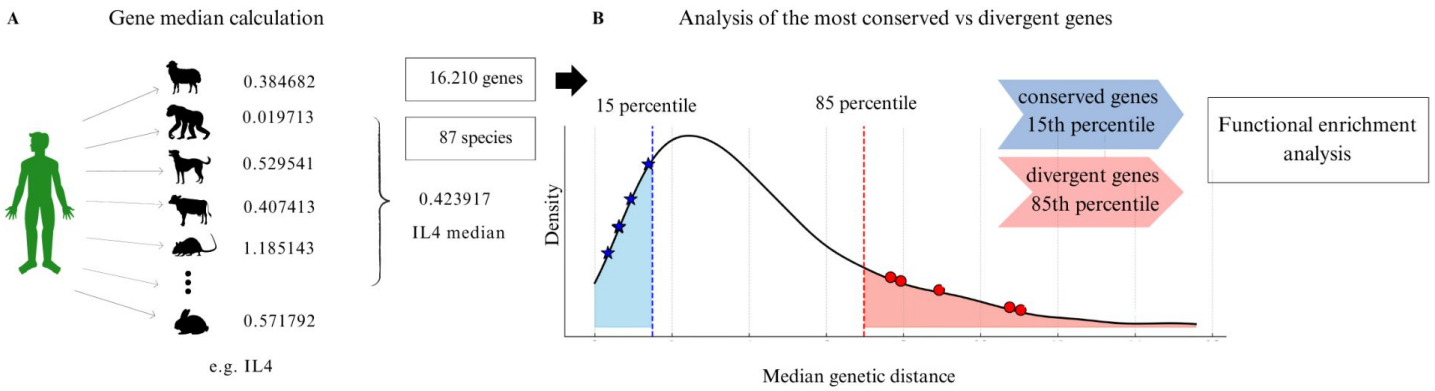


Figure 3: (A) Schematic illustration of how the median genetic distance was calculated for a gene, based on pairwise comparisons between *Homo sapiens* and other mammalian species. (B) Example of the distribution of median genetic distances across the 16,210 genes analyzed. Genes falling within the 15th percentile (most conserved) and the 85th percentile (most divergent) were selected for subsequent functional enrichment analyses.

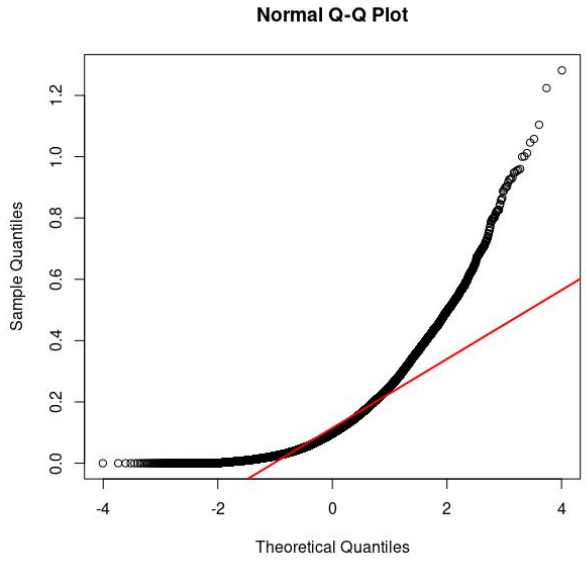


Figure 4: QQplot of the median evolutionary distances calculated for 16.210 single-copy orthologous genes between *Homo sapiens* and 87 mammalian species. The red line represents the reference for a standard normal distribution, and the divergence of the points indicates the distribution of the median evolutionary distances.

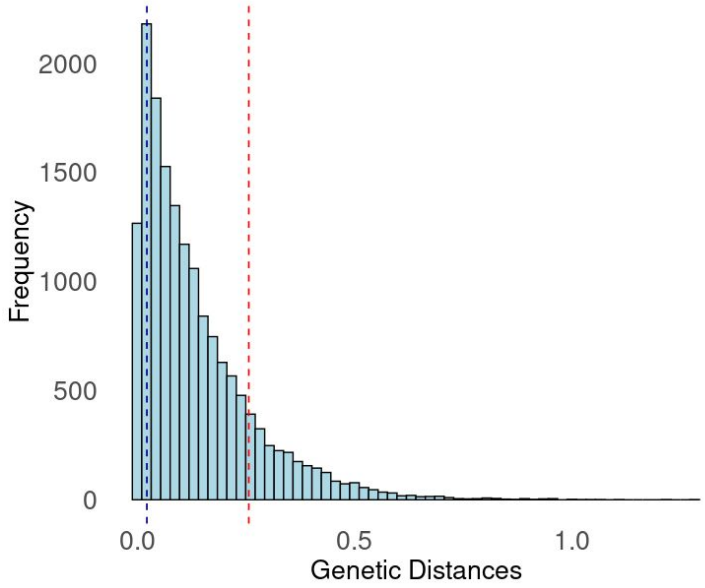


Figure 5: Histogram of genetic distances calculated for 16.210 single-copy orthologous genes between *Homo sapiens* and 87 mammalian species. The histogram illustrates the frequency of genetic distances, with a blue dashed line representing the 15th percentile and a red dashed line indicating the 85th percentile, marking thresholds for lower and higher divergence, respectively.

Functional enrichment analysis of genes at distribution extremes (15th and 85th percentiles)

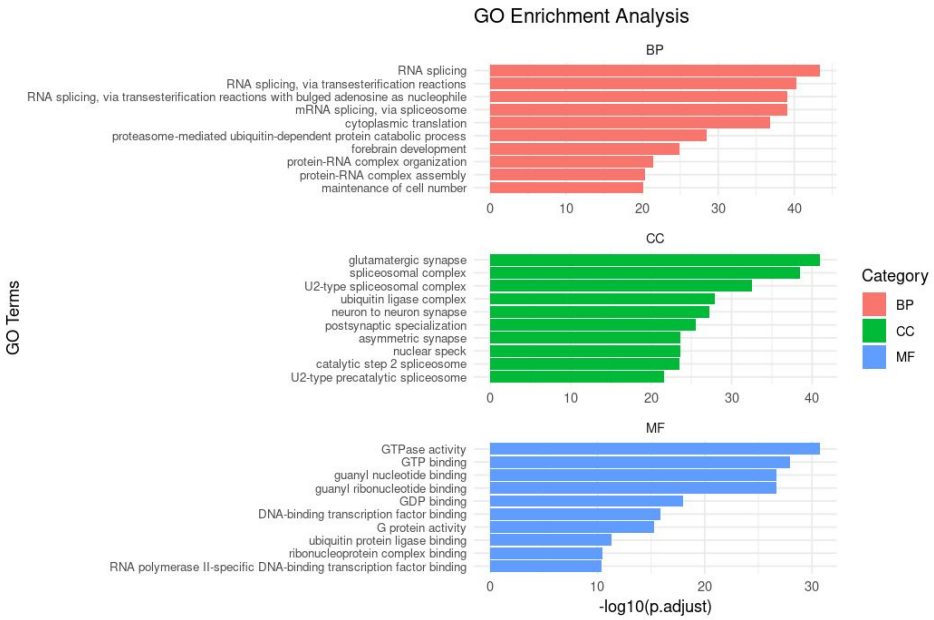


Figure 6: GO enrichment analysis for the biological process (BP), cellular component (CC) and molecular function (MF) categories of genes with medians within the 15th percentile in the distribution.

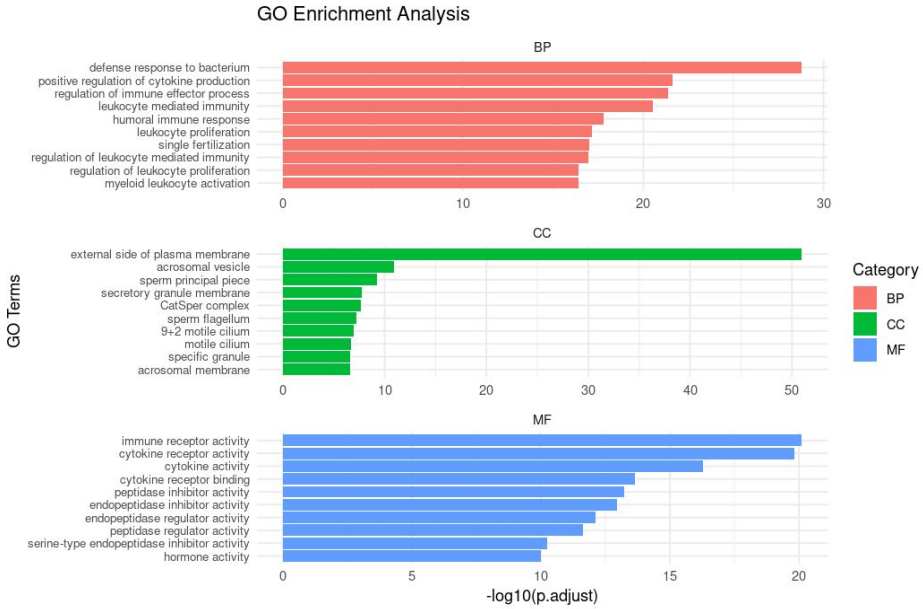


Figure 7: GO enrichment analysis for the categories of biological process (BP), cellular component (CC) and molecular function (MF) of genes with medians above the 85th percentile in the distribution.

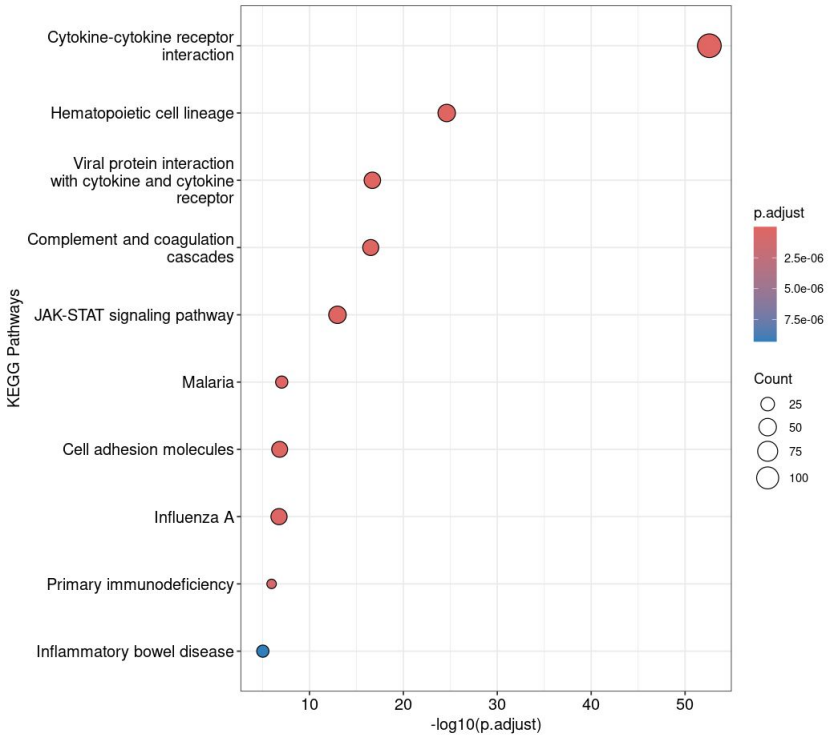


Figure 8: Pathway enrichment analysis using the KEGG database, focusing on genes in the 85th percentile of evolutionary distances.

Descriptive analysis of gene distances (comments):

Our results show that, consistent with literature, most proteins are highly conserved across mammals. This is reflected in a left-skewed distribution of median genetic distances, with values close to zero, indicating few average substitutions per site and highlighting essential biological processes that are maintained across species (KOONIN, 2008; KIMURA, 1983). In contrast, a smaller subset of genes showed higher divergence. Immune-related genes were notably overrepresented among the most divergent, displaying a greater number of substitutions per site over evolutionary time, suggesting that these genes are more evolutionarily dynamic.

Genes at the upper tail of the divergence distribution were enriched for biological processes such as “defense response to bacterium”, “positive regulation of cytokine production”, “regulation of immune effector process”, “leukocyte mediated immunity,” “regulation of leukocyte proliferation” and “myeloid leukocyte activation”. In terms of cellular localization, these genes were mostly found on the external side of the plasma membrane. Their molecular functions included immune receptor activity, cytokine activity, and regulation of peptidase activity, indicating that cytokines, their receptors, and other immune receptors are among the most rapidly evolving genes. KEGG pathway analysis confirmed enrichment in “cytokine-cytokine receptor interaction,” followed by “hematopoietic cell lineage,” “viral protein interaction with cytokine and cytokine receptor,” and “complement and coagulation cascades”. Our findings align with the understanding that genes involved in host-pathogen interactions are evolving faster due to strong positive selective pressures, leading to rapid evolution to keep pace with evolving pathogens.

In addition, some reproductive processes were also enriched among the most divergent genes, such as “single fertilization,” “acrosomal vesicle,” and “sperm flagellum.” This finding supports previous studies reporting rapid evolution of immune and reproductive genes across species due to host-pathogen coevolution and sexual selection pressures (SWANSON; VACQUIER, 2002; NIELSEN et al., 2005).

In contrast, the most conserved genes, within the 15th percentile of median evolutionary distances, were involved in fundamental biological processes essential to all mammals. These included "RNA splicing," "cytoplasmic translation," "proteasome-mediated ubiquitin-dependent protein catabolic process," and "maintenance of cell number." Cellular components associated with these genes were "glutamatergic synapse" and "spliceosome," while molecular functions involved "GTPase activity," "DNA binding," and "transcription factor activity." The conservation of these genes reflects their critical roles in basic cellular functions, where alterations can be deleterious.

Immune Pathways

To investigate evolutionary patterns in immune-related gene sets, two complementary approaches were applied:

1. **Overall divergence of immune pathways**: an exploratory analysis to estimate the typical divergence of each pathway across all 87 mammalian species. Using the median genetic distances of each gene, we computed the median distance for each immune pathway in the KEGG and Reactome databases based on its constituent genes.

2. **Species-specific divergence:** The second strategy focused on identifying species-specific divergence signals. For this, genes were ranked by their genetic distance to the human sequence in each species, and a Gene Set Enrichment Analysis (GSEA) was performed independently for each species to identify pathways enriched among the most divergent genes.

1. Overall divergence of immune pathways:

Pathway	Median	IQR	Coverage
Antimicrobial peptides	0.418514	0.253	52%
Interleukin 36 pathway	0.409024	0.303	100%
Interleukin 18 signaling	0.379585	0.073	87,50%
Dectin 2 family	0.362369	0.338	69,20%
Interleukin 1 processing	0.359283	0.333	88%
Tnfs bind their physiological receptors	0.339076	0.191	100%
Trif mediated programmed cell death	0.325407	0.071	100%
Tlr3 mediated ticam1 dependent programmed cell death	0.325093	0.097	100%
Interleukin 10 signaling	0.320919	0.186	78%

Table 1: Top 10 Reactome pathways with highest estimated evolutionary distances

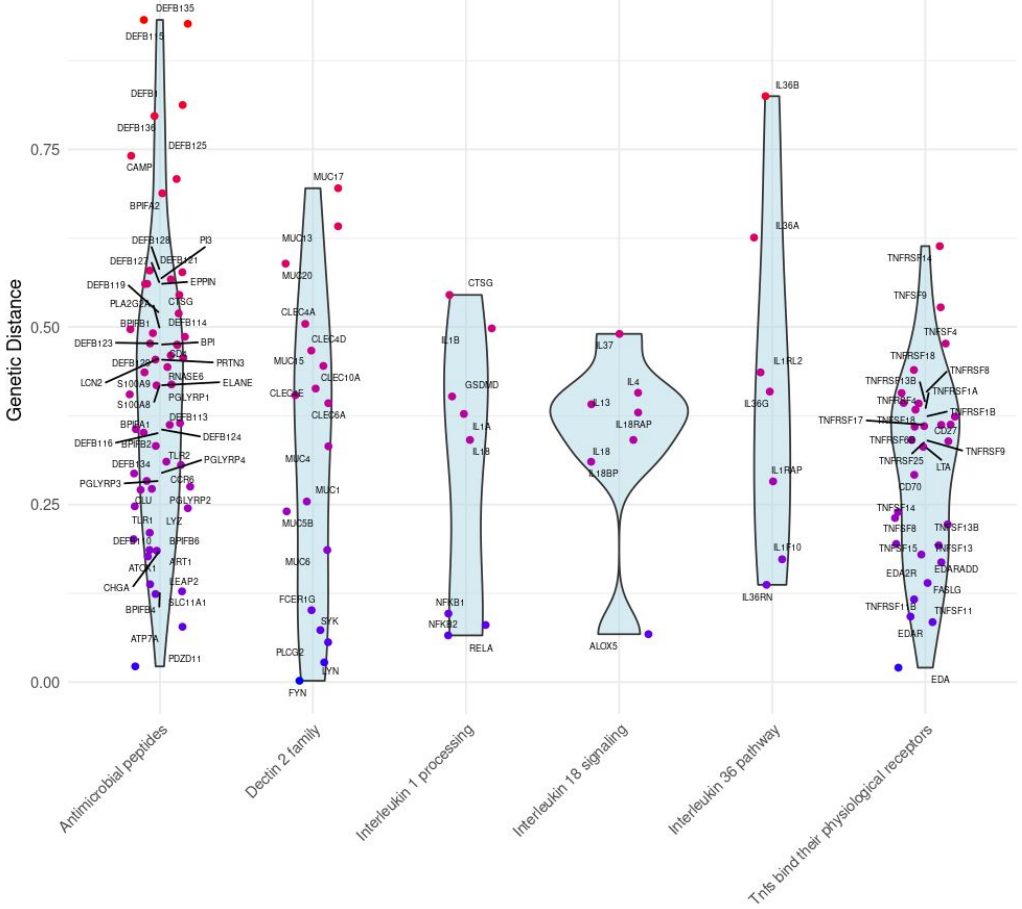


Figure 9: Violin plots of Reactome immunological pathways (Antimicrobial peptides; Dectin 2 family; Interleukin 1 processing; Interleukin 18 signaling; interleukin 36 pathway; Tnfs bind their physiological receptors), illustrating the variability of median evolutionary distances between constituent genes of the pathway.

After estimating overall divergence using the median genetic distances of genes, the GSEA approach allowed comparing species-specific differences. The analysis was conducted separately for each of the 87 mammalian species, ranking genes by their genetic distance to *Homo sapiens*. Of the 1,097 Reactome pathways tested, 283 were significantly enriched ($q \leq 0.05$) in at least one species. The number of significant pathways varied among species, but immune-related processes consistently ranked among the most recurrent categories (figure 10). The pathways most frequently enriched with divergent genes (highest genetic distances values) included **immunoregulatory interactions between lymphoid and non-lymphoid cells, the innate immune system, antimicrobial peptides, cytokine signaling in the immune system, interleukin signaling, neutrophil degranulation, costimulation by the CD28 family, interferon signaling, and the complement cascade.**

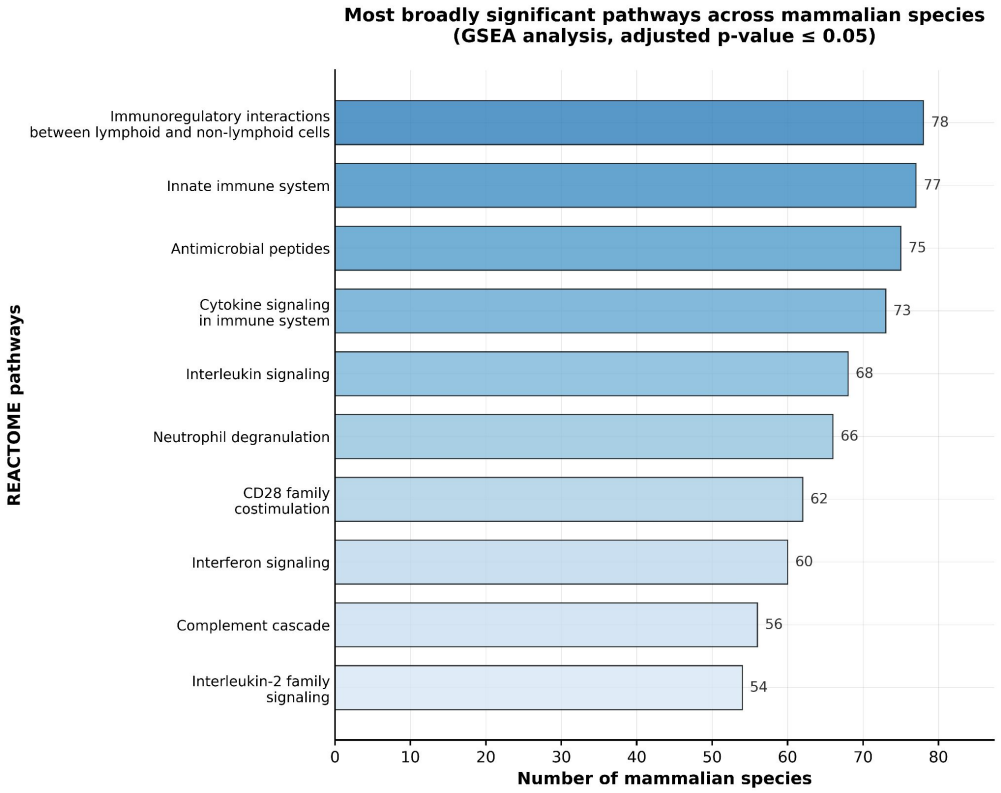


Figure 10: Top 10 Reactome pathways most broadly enriched with highly divergent genes across 87 mammalian species, as identified by GSEA. Bars show the number of species in which each pathway is statistically significant (adjusted p-value ≤ 0.05). Immune-related processes were among the most recurrently enriched pathways, highlighting consistent evolutionary divergence.

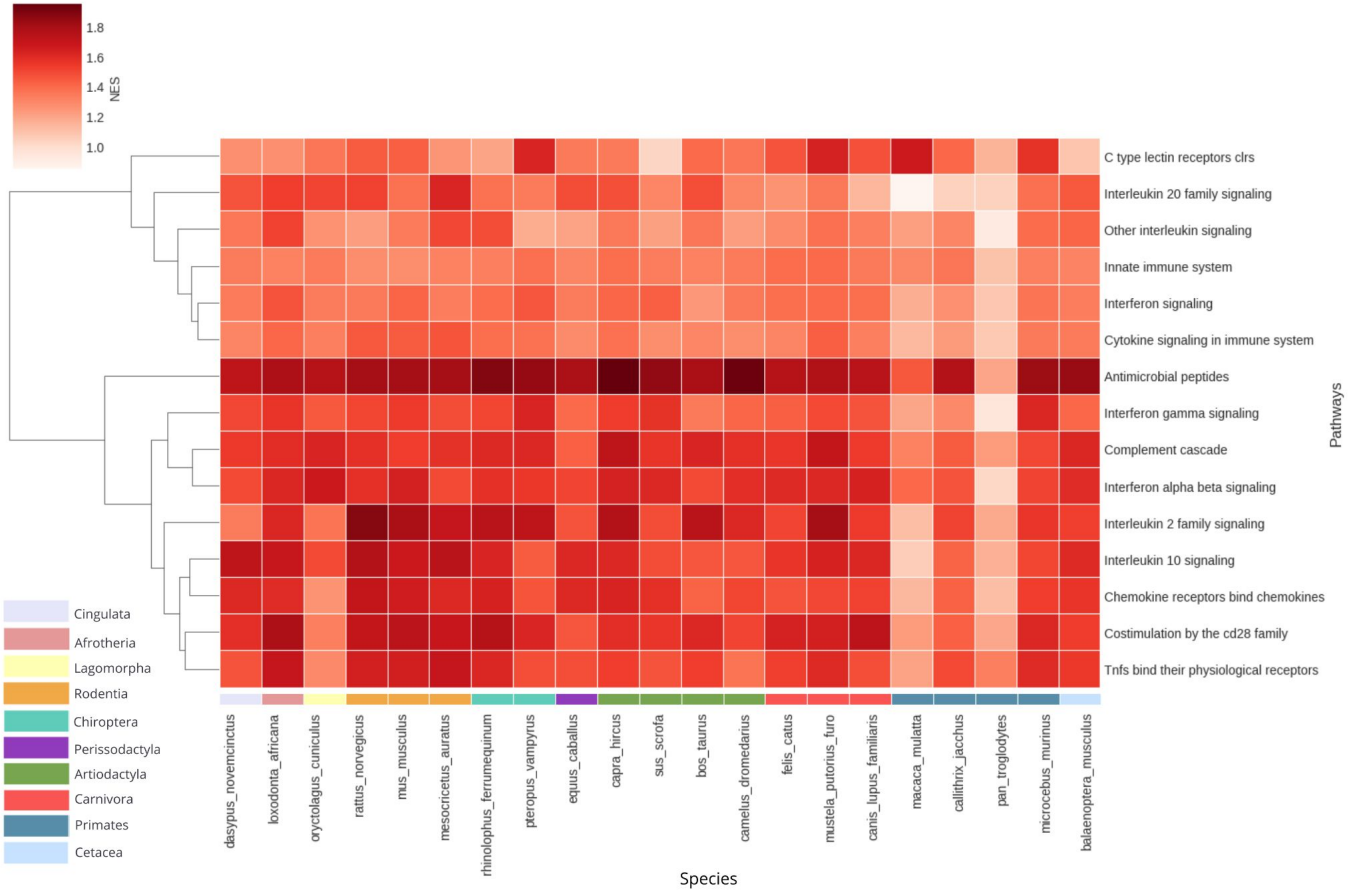
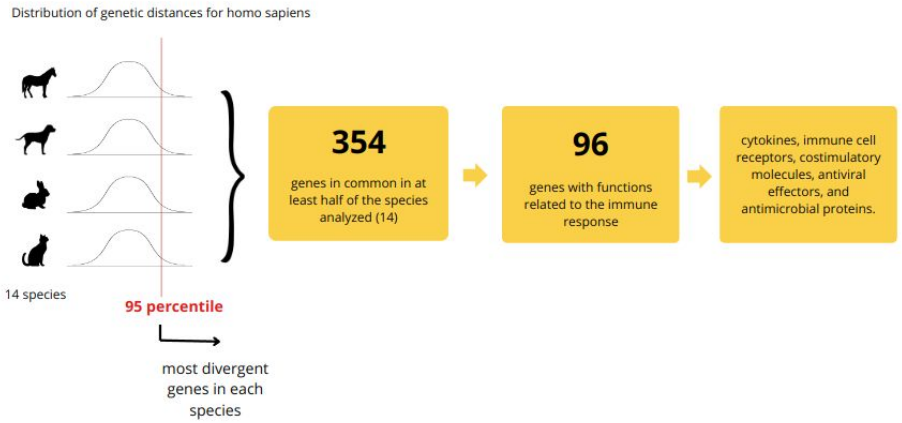


Figure 11: Heatmap of NES for Reactome pathways across a subset of mammalian species. Each cell represents the NES value of a given pathway (rows) in a given species (columns), with darker shades indicating stronger enrichment. Species are grouped by taxonomic order (color-coded bars at the bottom), and pathways are clustered based on similarity in enrichment patterns. To improve visualization, only the pathways with the highest NES values and a selection of species are shown.

High divergence immune genes

From the total of 87 species analyzed, 14 were selected for focused analyses *(Callithrix jacchus, Macaca mulatta, Pan troglodytes, Bos taurus, Sus scrofa, Capra hircus, Equus caballus, Canis lupus familiaris, Felis catus, Oryctolagus cuniculus, Mesocricetus auratus, Mus musculus, Rattus norvegicus, and Rhinolophus ferrumequinum*). These species were chosen for their higher genome quality and coverage while maintaining broad phylogenetic diversity. Within this set of species, we identified genes that consistently ranked above the 95th percentile of genetic distance to *Homo sapiens* in at least half of the species (≥ 7), representing a set of the most divergent genes. This analysis yielded 354 recurrently divergent genes, 94 of which were annotated as immune-related. The heatmap (figure 12) summarizes examples of these genes (ordered by most divergent) and their estimated distances across species.



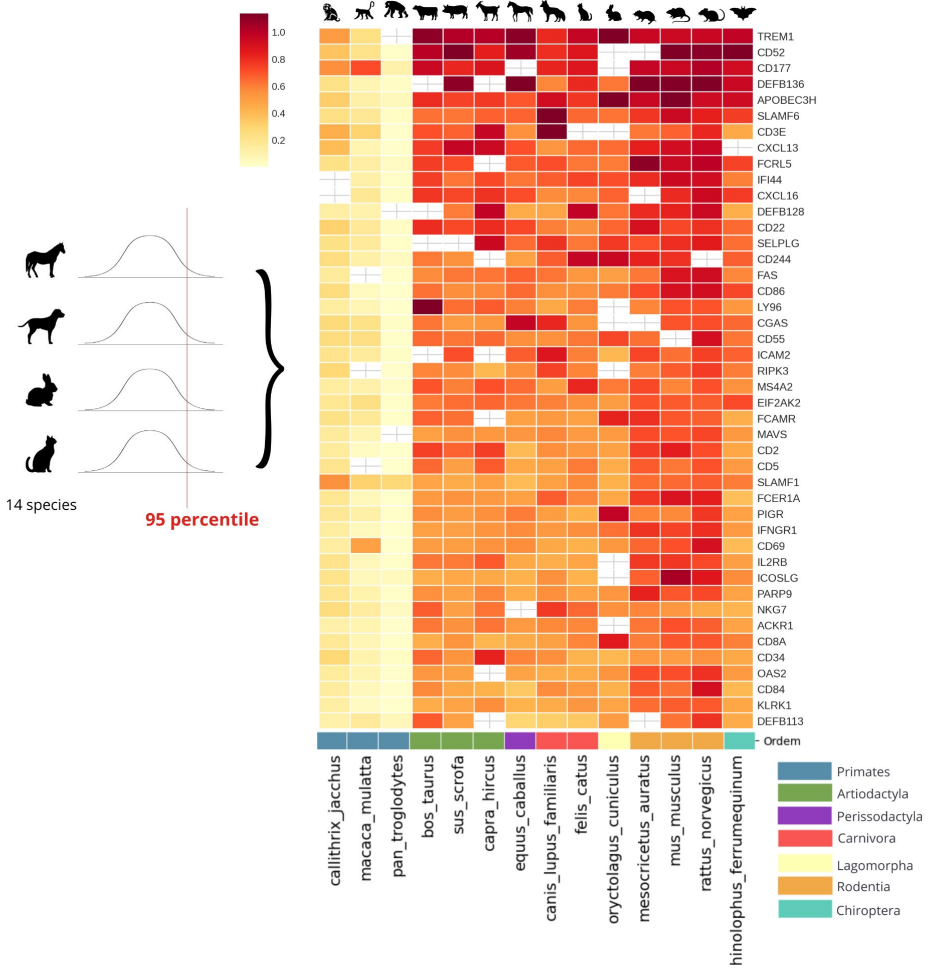


Figure 12: Heatmap of the most divergent immune-related genes across 14 mammalian species. Warmer colors (red) indicate greater divergence and lighter colors (yellow) indicating lower divergence. White cells represent missing data for specific gene-species comparisons.

Analysis of Non-synonymous to Synonymous Substitution Ratios (dN/dS)

Genes showing high genetic distance estimates between species often accumulated a greater number of amino acid-changing substitutions. Such divergence can arise from neutral processes, but it may reflect episodes of positive selection, where adaptive changes in immune genes could be favored due to host-pathogen interactions. To investigate this possibility, we applied codon-based models to identify genes and specific sites evolving under positive selection. Our analysis aimed to detect which genes show evidence of adaptive evolution and quantify the number of positively selected sites within them. In addition, **different gene sets were compared according to different criterias:**

The candidate gene set was defined using the following approaches:

1. **Pathway-driven set:** In the GSEA results, the Reactome pathway ***Interferon Signaling*** was among the highest normalized enrichment scores (NES). Given its central role in antiviral defense and the interspecific variation in interferon pathways and antiviral outcomes (GONZALEZ et al., 2025; ZHOU et al., 2016), this pathway was selected for targeted analysis. All genes annotated in Reactome as belonging to *Interferon Signaling* were selected; from these, only one-to-one orthologs with reliable sequence data and genetic distance estimates were retained. This resulted in a final set of 190 genes included in the codeml analyses.

2. **Cross-species high-divergence set:** For each of the 14 species (*Callithrix jacchus, Macaca mulatta, Pan troglodytes*, *Bos taurus, Sus scrofa, Capra hircus*, *Equus caballus*, *Canis lupus familiaris, Felis catus*, *Oryctolagus cuniculus*, *Mesocricetus auratus, Mus musculus, Rattus norvegicus*, *Rhinolophus ferrumequinum*), genes were ranked by their genetic distance from *Homo sapiens* and those in the 95th percentile were selected. Genes that appeared in at least half of the species (≥ 7 of 14) were retained, totalling 354 genes. Functional annotations from GO (BOTSTEIN et al., 2011) and GeneCards (SAFRAN et al., 2010) were used to retain only immune-related genes. This filtering yielded 94 candidates, which were then tested for positive selection with codeml.

For comparison, two additional gene sets were included alongside the interferon and high-divergence immune groups, to contrast patterns of positive selection in highly conserved genes and in a neutral random sample:

3. **Cross-species conserved set:** Using the same approach, for each of the 14 species genes in the 5th percentile of genetic distances to *Homo sapiens* were identified. Genes consistently falling in this lower tail in at least half of the species (≥ 7 of 14) were retained as conserved candidates. From this pool, 92 genes were randomly selected to generate a set of highly conserved genes for comparative analyses.

4. **Random set:** As a neutral reference, 87 genes were randomly sampled from the complete pool of 16,210 one-to-one orthologs.

Together, these four gene sets (190 interferon, 94 high-divergence immune, 92 conserved, and 87 random) were analyzed with codeml and used for comparative enrichment analyses of positive selection. Statistical comparisons between sets were conducted using Fisher's exact test. Because not all 14 species had one-to-one orthologs for every gene, the number of species' sequences analyzed varied among genes.

Among the 94 highly divergent immune genes (gene median above 95th percentile), **63.8%** (60/94) met these criteria, compared with **16.3%** (31/190) of the interferon signaling genes. **None** of the conserved genes (gene median under 5th percentile) showed evidence of selection, while **13.4%** (11/82) of random genes did. Fisher's exact tests confirmed a strong enrichment of positive selection in divergent immune genes compared with both conserved and random sets (Fisher's test p value $p < 10^{-23}$ and $p < 10^{-11}$, respectively), and in interferon signaling genes compared with conserved ones (Fisher's test p value $p < 10^{-6}$). In contrast, no significant difference was observed between the interferon set and the random sample (Fisher's test p value = 0.47) (figure 13).

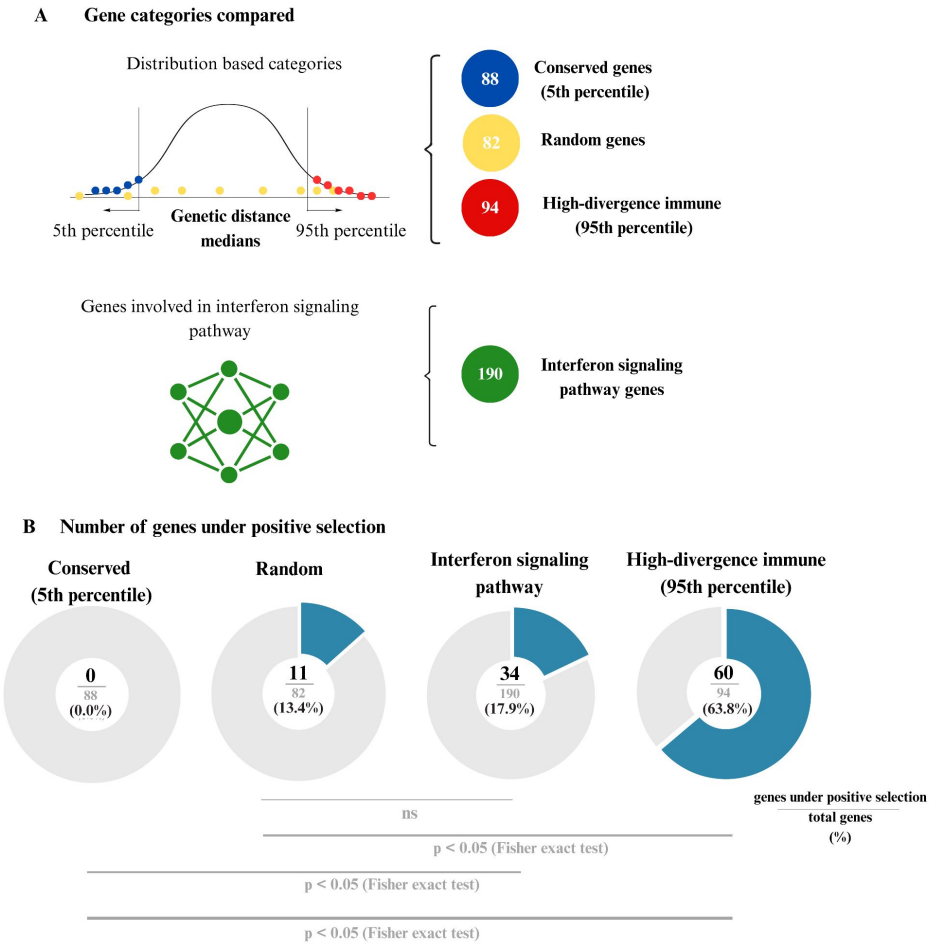


Figure 13: (A) Gene categories analyzed: conserved genes (below 5th percentile of genetic distance), random genes (sampled from the full ortholog set), highly divergent immune genes (above 95th percentile), and genes from the Reactome interferon signaling pathway. (B) Proportion of genes under positive selection in each category, as determined by codeml (M7 vs. M8, $q \leq 0.05$; BEB ≥ 0.95). Percentages are shown inside the charts, with absolute counts (selected/total). Statistical comparisons between groups were performed using Fisher's exact test, with significant differences indicated below the plots.

Most significant genes contained only a few positively selected sites (between 1 to 5). Within the set of highly divergent immune genes (95th percentile), the strongest signals were detected in **PARP9** (42 sites), **EIF2AK2** (32 sites), and **IFI44** (30 sites). Additional immune genes with multiple positively selected sites included CD22 (18), CD244 (16), TSLP (14), CD3E (13), and FCER1A (13) (figure 14).

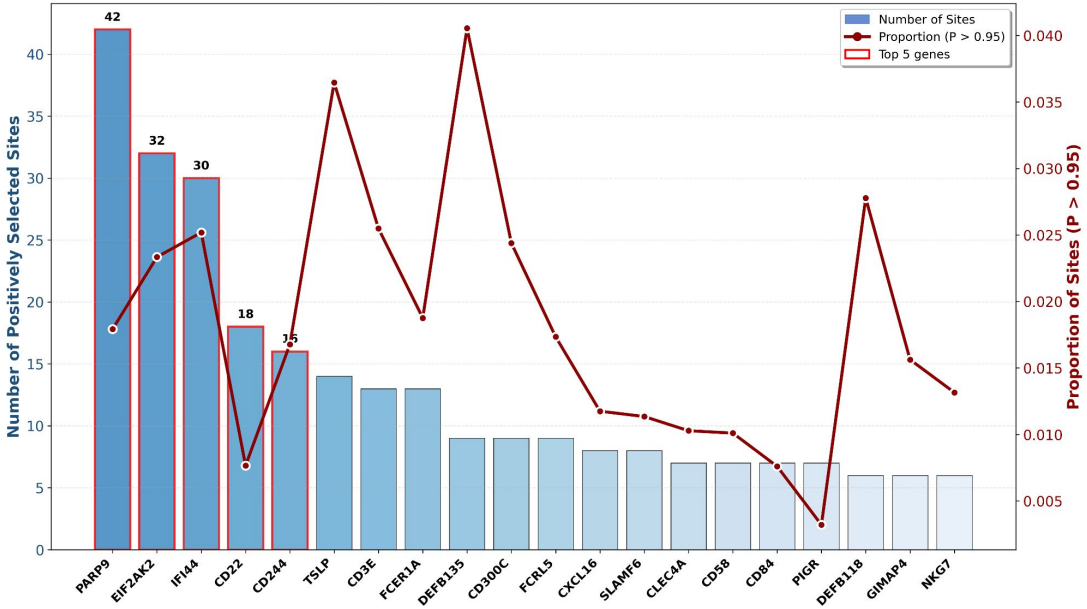


Figure 14: Positive selection analysis of the cross-species high-divergence set, ranked by the number of positively selected sites. Bars (left axis) shows the number of sites with BEB (≥ 0.95), while the line (right axis) indicates the proportion of sites with positive selection relative to the total number of sites.

In the interferon signaling pathway, the genes with the highest number of sites under positive selection are **EIF2AK2** (32 sites), **OAS1** (15 sites), and **RNASEL** (12 sites) (figure 15).

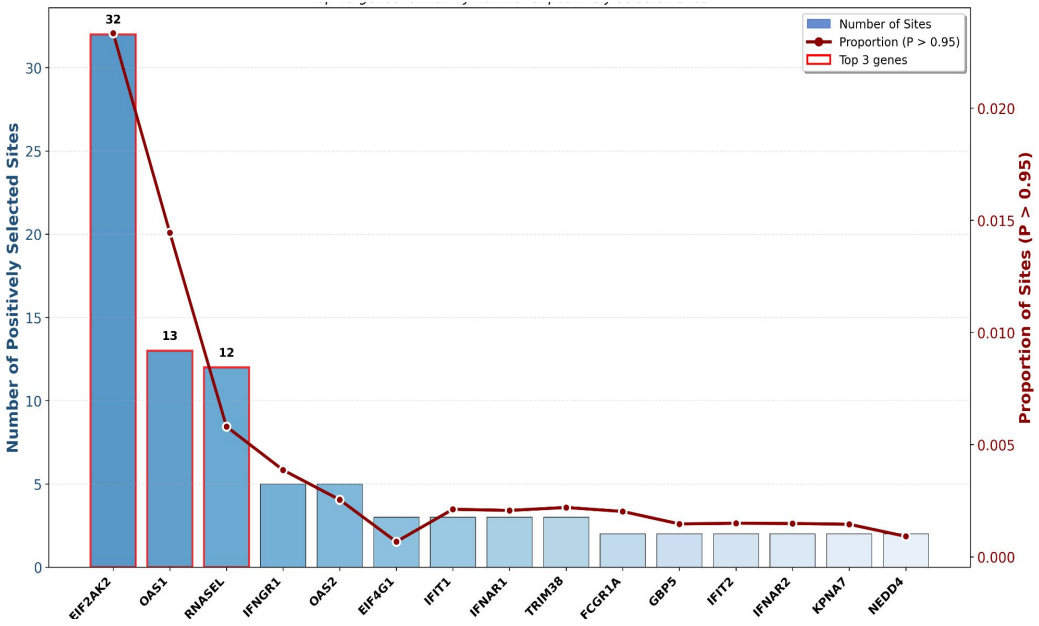


Figure 15: Positive selection analysis of the set 20 immune genes from the interferon signaling pathway set, ranked by the number of positively selected sites. Bars (left axis) shows the number of sites with BEB (≥ 0.95), while the line (right axis) indicates the proportion of sites with positive selection relative to the total number of sites.

PPI Network analysis

When projecting the 190 genes of the interferon signalling pathway onto a **protein-protein interaction network**, the genes clustered into five modules, identified by community detection (GLay clustering), representing groups with stronger internal connectivity. The network allowed us to integrate information about the genetic distance and the signals of adaptive evolution of each gene, allowing to identify groups of genes that accumulated the greatest divergence and showed signals of positive selection (figure 16) and investigate their relationships in the network. We further compared, among genes with evidence of selection, the number of sites with BEB $\geq$ 0.95 (figure 17).

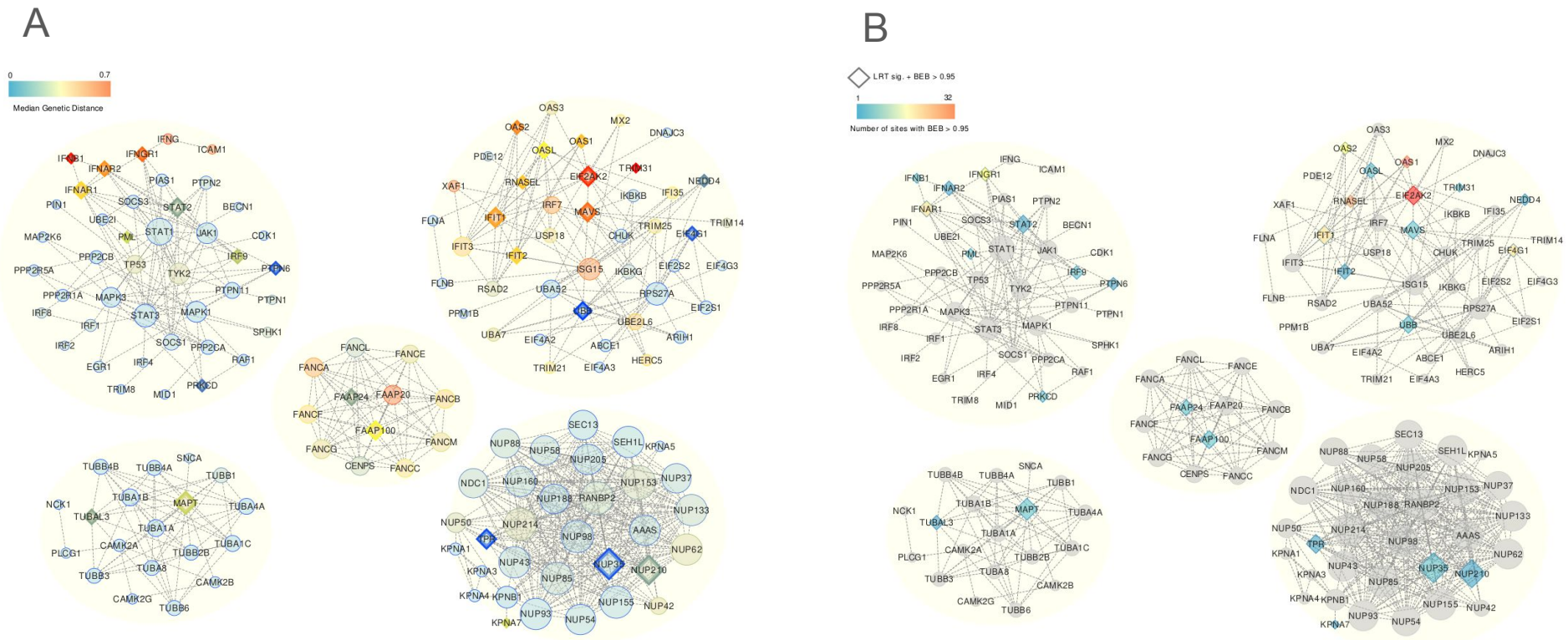


Figure 16: Protein-protein interaction (PPI) network of genes in the Reactome interferon signaling pathway.

A) Node size indicates connectivity, **node color represents the median genetic distance to *Homo sapiens*** across 87 species (section 2.2.2), and diamonds shape mark genes with evidence of positive selection (LRT M7 vs. M8, $q \leq 0.05$, BEB ≥ 0.95).

B) Node size indicates connectivity, diamonds mark genes with evidence of positive selection and **node color represents the number of positively selected sites** (BEB ≥ 0.95). Gray nodes represent genes without evidence of positive selection in this study.

PPI Network analysis (comments)

Among all the modules in the resulting network, the module enriched in antiviral effectors, including EIF2AK2, TRIMs, OAS genes (OAS1, OAS2, OASL), RNASEL, and IFITs, stands out in terms of the **number of sites under positive selection** and the **greatest median genetic distance**. Within this cluster, EIF2AK2 is particularly notable for its high connectivity within the module, being the gene with the highest number of positively selected sites and higher median genetic distance. Furthermore, interferon receptors such as IFNAR1, IFNAR2, and IFNGR1 also have signals of positive selection, but with a low number of sites (1 to 5 sites). In contrast, pathway regulators, including STAT1, JAK1, MAKP1, MAPK3, and IRF9, emerged as highly connected hubs that remain conserved, showing no evidence of positive selection. Genes involved in nuclear transport, such as the NUP family, also formed a large and densely connected cluster, but were largely conserved. Finally, some genes outside the canonical antiviral core, such as members of the Fanconi anemia complex (FANC) and tubulin-related genes (TUB), also show signs of positive selection, but limited to a single site with BEB $\geq$ 0.95.

Detection of Branch-Specific Positive Selection

To investigate whether positive selection occurred in specific species, we performed branch-site analysis with *aBSREL** on two genes: **PARP9** and **EIF2AK2**, which concentrate the **highest number of sites** under positive selection.

*Adaptive Branch-Site Random Effects Likelihood (aBSREL) method implemented in HyPhy (SMITH et al., 2015), which detects episodic positive selection acting on individual branches of a phylogenetic tree.

Evidence of selection in **PARP9** was significant in **6 of the 14 species**: cow (*Bos taurus*), bat (*Rhinolophus ferrumequinum*), dog (*Canis lupus familiaris*), horse (*Equus caballus*), marmoset (*Callithrix jacchus*), and rat (*Rattus norvegicus*) (figure 18)

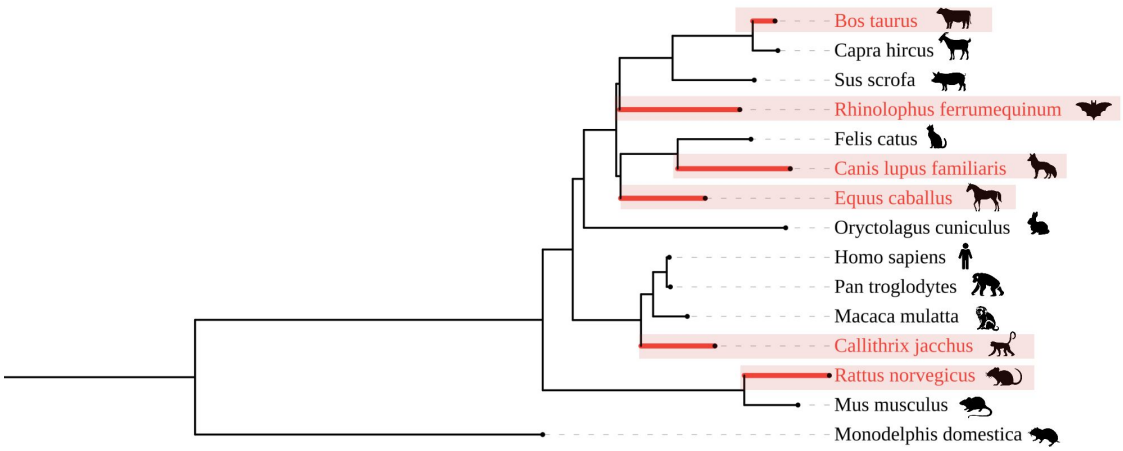


Figure 18: Phylogenetic tree of the PARP9 gene across 14 mammalian species analyzed**.** The topology was inferred from codon-based alignments (as described in 2.3.3). Branches highlighted in red indicate lineages with significant evidence of episodic positive selection, as detected by the aBSREL method (adjusted p-value ≤ 0.05). *Monodelphis domestica* (marsupial) was included as an outgroup for visualization purposes only and was not considered in the selection analysis.

For **EIF2AK2**, most species showed evidence of positive selection, with significance detected in **11 of the 15 species** analyzed, except in cow (*Bos taurus*), pig (*Sus scrofa*), dog (*Canis lupus familiaris*), and chimpanzee (*Pan troglodytes*) (figure 19). Therefore, the results show that among the species analyzed, the EIF2AK2 gene presents evidence of positive selective pressure in **all taxonomic orders** analyzed.

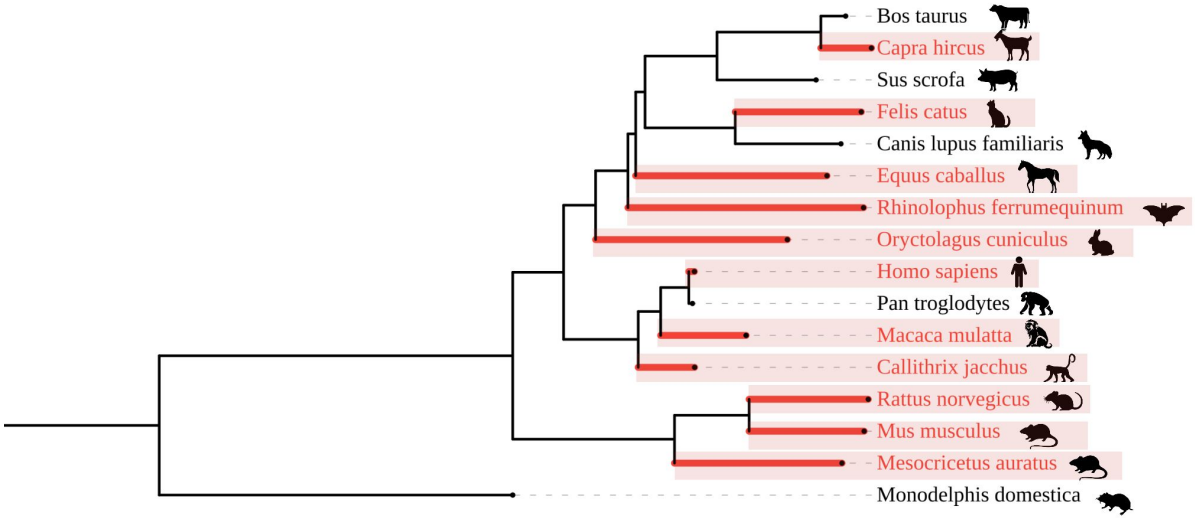


Figure 19: Phylogenetic tree of the EIF2AK2 (PKR) gene across 15 mammalian species analyzed**.** The topology was inferred from codon-based alignments (as described in 2.3.3). Branches highlighted in red indicate lineages with significant evidence of episodic positive selection, as detected by the aBSREL method (adjusted p-value ≤ 0.05). *Monodelphis domestica* (marsupial) was included as an outgroup for visualization purposes only and was not considered in the selection analysis.

To compare patterns of interspecific divergence, we constructed a phylogenetic tree of canonical EIF2AK2 protein sequences among mammals, with distances to *Homo sapiens* represented in bar graphs (Figure 20).

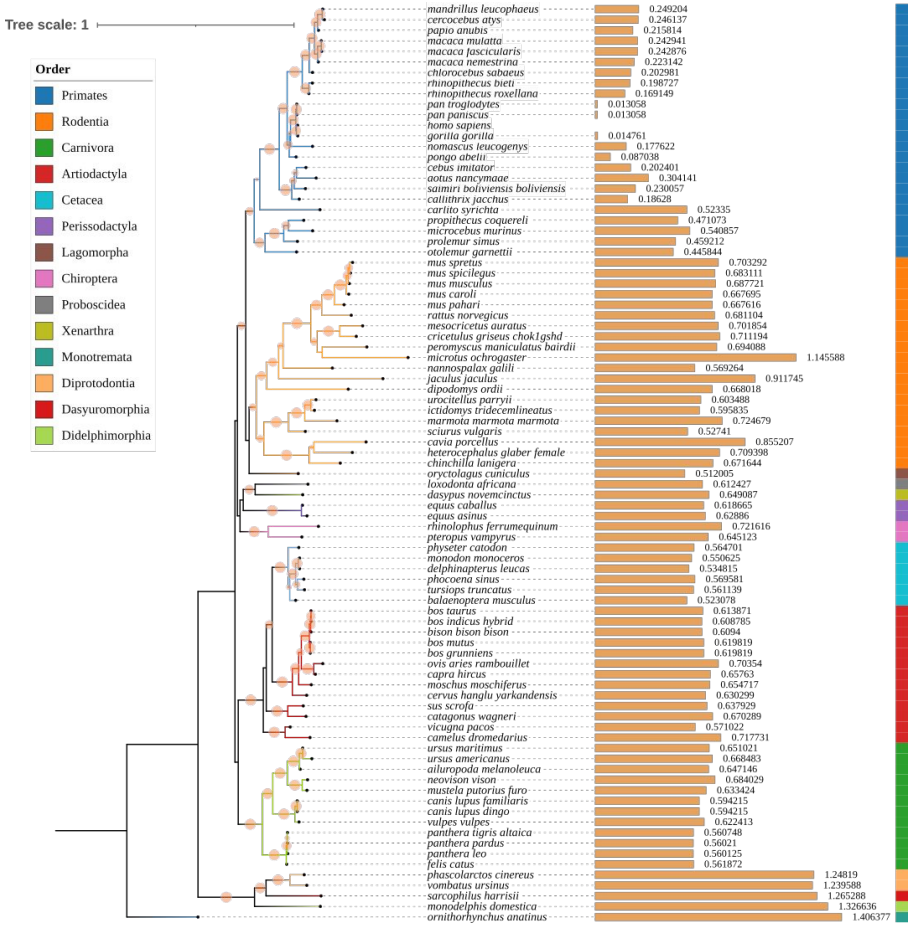


Figure 20: Phylogenetic tree of the EIF2AK2 gene based on canonical protein sequences from mammalian species. The topology was inferred using the maximum likelihood method implemented in RAxML PROTGAMMAAUTO model selection, 1000 rapid bootstrap replicates), and the tree was rooted with the platypus (*Ornithorhynchus anatinus*). Filled circles at internal nodes indicate RAxML bootstrap support; circle size/shade increases with support (small/light = 50-79%, medium = 80-94%, large/dark ≥ 95%); nodes < 50% have no circle. Side bars represent the genetic distances of each species to *Homo sapiens*, expressed as the average number of substitutions per site. Branch colors correspond to the taxonomic order of each species. Tree annotation was done using iTOL software (LETUNIC; BORK, 2007).

Species-level regression analysis

To investigate if some species accumulate substitutions more strongly than others as genetic divergence increases, we regressed relative divergence (Δ) on overall gene divergence for each species and compared the resulting slopes (β). The regression slope (β) captures how strongly each species' divergence from humans increases as genes become more divergent, reflecting lineage-specific patterns of substitution accumulation. This allows comparison of divergence patterns across species and taxonomic orders.

Overall divergence for each gene was defined as the median genetic distance to *Homo sapiens* across species. We standardized it across genes using z-scores (subtracting the mean across genes and dividing by the standard deviation).

Relative divergence was defined, for gene *g* and species *s*, as:

$$\triangle_{g,s} = d_g(s, H) - d_g(C, H)$$

d = evolutionary distance
g = gene
S = species
H = *Homo sapiens*
C = Chinpanzee

This metric quantifies how much more (or less) divergent a species is from humans compared to the chimpanzee. We used *P. troglodytes* as the reference due to its close evolutionary relationship to humans, providing a stable baseline.

The analysis was performed on **two gene sets**: the complete set of 15,093 genes with available data for *Pan troglodytes* and a subset of 2,205 genes annotated to the "immune system process" category of the GO (GO:0002376). For each of the 86 mammalian species, we computed two regression slopes (β): based on the full gene set (β all) and only on immune-related genes (β immune). To test if immune genes to show higher slopes than the complete set of genes, we compared β immune and β all values within each species using a one-sided paired Wilcoxon signed-rank test (H1: β immune > β all).

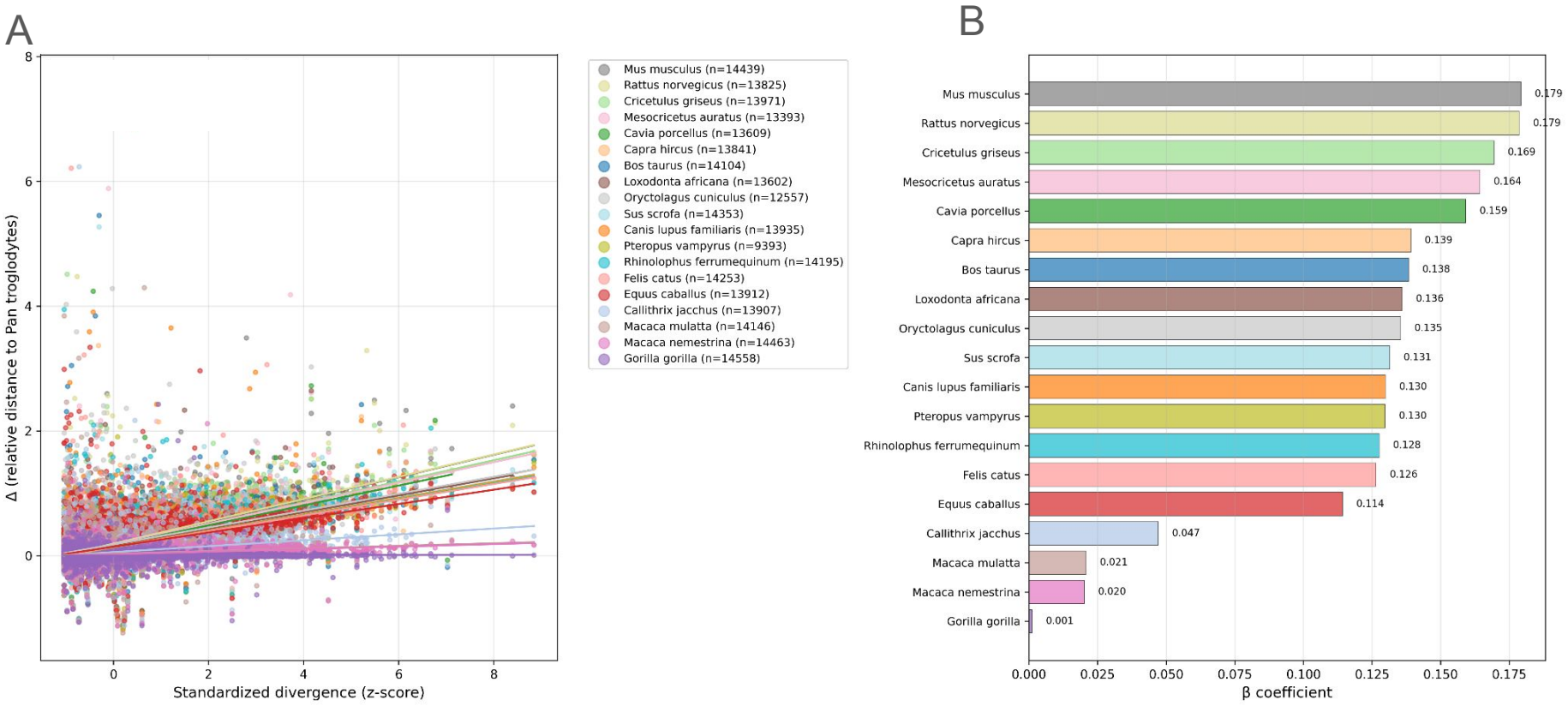


Figure 21. Species-specific slopes of relative divergence (Δ) vs. global gene divergence (15,093 genes). (A) Scatter plot showing the relationship between gene-level global divergence (x-axis, standardized as z-scores of median genetic distance across species) and the relative distance to *Homo sapiens* (y-axis), expressed as Δ = d(species, *Homo sapiens*) - d(*Pan troglodytes*, *Homo sapiens*). Each point represents a gene, with colors indicating species. Linear regressions were fitted separately for each species. The inset summarizes the total number of observations, the number of species, the mean slope (β), and the overall fit (R^2). (B) Ranking of species-specific β coefficients. The β coefficient corresponds to the slope of the regression line, reflecting how strongly the relative distance to *Homo sapiens* increases with gene divergence. Higher β values indicate that a species has a tendency to accumulate more substitutions in genes that are globally more divergent, whereas lower β values indicate weaker sensitivity to divergence.

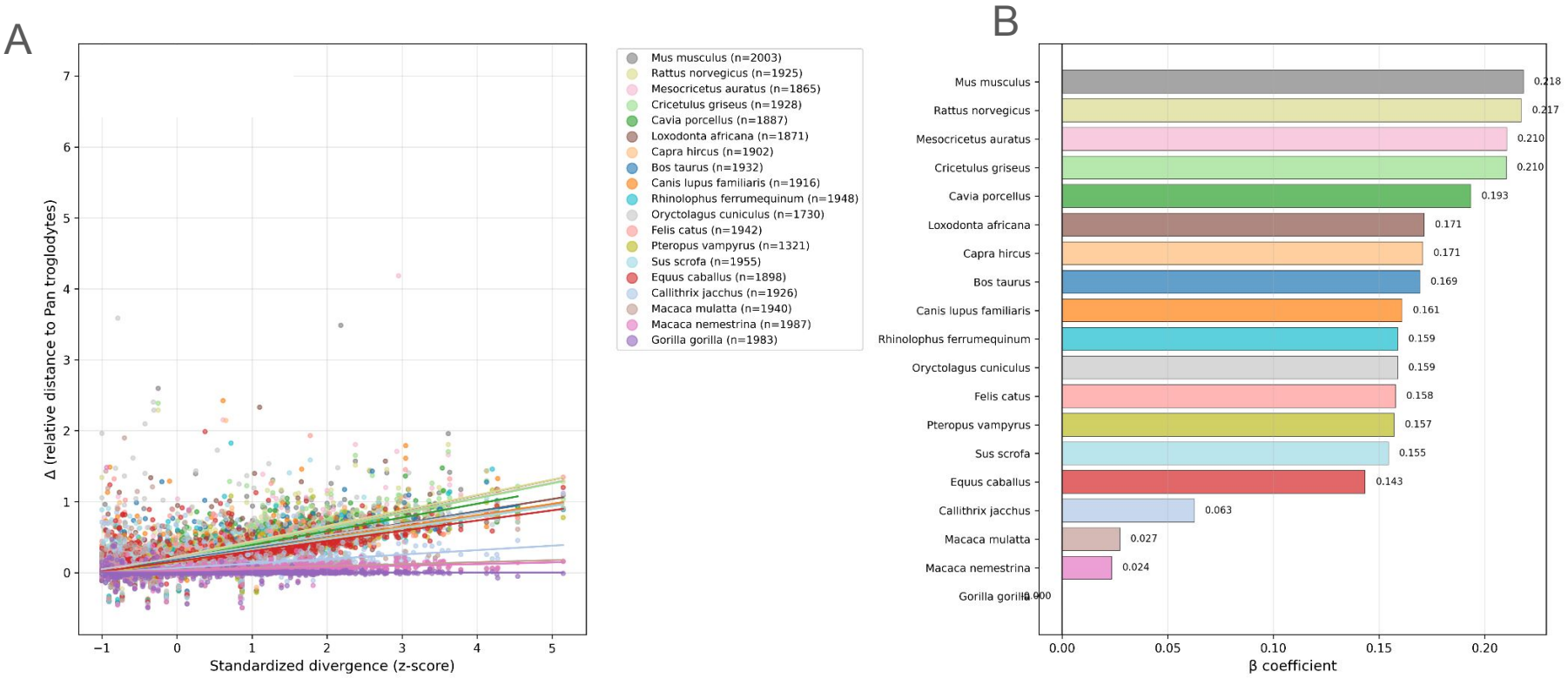


Figure 22. Species-specific slopes of relative divergence (Δ) vs. global gene divergence, across 2,205 genes related to the GO term immune system process (GO:0002376). (A) Scatter plot showing the relationship between gene-level global divergence (x-axis, standardized as z-scores of median genetic distance across species) and the relative distance to *Homo sapiens* (y-axis), expressed as Δ = d(species, *Homo sapiens*) - d(*Pan troglodytes*, *Homo sapiens*). Each point represents a gene, with colors indicating species. Linear regressions were fitted separately for each species. The inset summarizes the total number of observations, the number of species, the mean slope (β), and the overall fit (R^2). (B) Ranking of species-specific β coefficients. The β coefficient corresponds to the slope of the regression line, reflecting how strongly the relative distance to *Homo sapiens* increases with gene divergence. Higher β values indicate that a species has a tendency to accumulate more substitutions in genes that are globally more divergent, whereas lower β values indicate weaker sensitivity to divergence.

Species-level regression analysis

In both scenarios, the ranking of species according to the largest β coefficients remained similar, with the rodent group maintaining the highest values, particularly mice (*Mus musculus*) and rats (*Rattus norvegicus*), followed by intermediate species represented by ungulates (*Capra hircus, Bos taurus, Sus scrofa*), elephants (*Loxodonta africana*), rabbits (*Oryctalagus cuniculus*), carnivores (*Canis lupus and Felis catus*), and bats (*Pteropus vampyrus, Rhinolophus ferrumequinum*). The horse (*Equus caballus*) remained as the non primate species with the lowest β.

The slope was observed to be greater in immunological genes (mean $\beta = 0.143$) than in the complete set (mean $\beta = 0.116$). To test whether immune related genes have higher species specific slopes than the complete set of genes, we compared the β values obtained from the immune genes of each species with those of the complete gene set using a Wilcoxon paired-rank test (H1: β immune > β all). Across the 86 species, the test was significant ($W = 3736.0$, $p = 4.76 \times 10^{-16}$), with a positive median paired difference ($\Delta\beta = 0.0281$). This result demonstrates that, across species, immune related genes accumulate substitutions more strongly across species as genetic divergence increases.